

Universidade de Brasília
Faculdade de Tecnologia
Departamento de Engenharia Civil e Ambiental

Aplicação do método da rigidez direta

Trabalho I

Disciplina: Teoria das Estruturas II

Tarso de Matos Bessa (22/2013950)

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1. Dados

Para o aluno deste trabalho, foram atribuídos os seguintes dados:

a	2,0 m
b	2,0 m
c	3,0 m
P	10 kN
q	0,5 kN/m
M	2 kN · m
E	180 GPa
ν	0,3
A	$5{,}0 \times 10^{-3}\text{m}^2$
I	$2{,}1 \times 10^{-6}\text{m}^4$
J	$3{,}5 \times 10^{-6}\text{m}^4$

2. Exercício I

Determinar os deslocamentos nodais e as forças axiais nas barras da treliça abaixo:

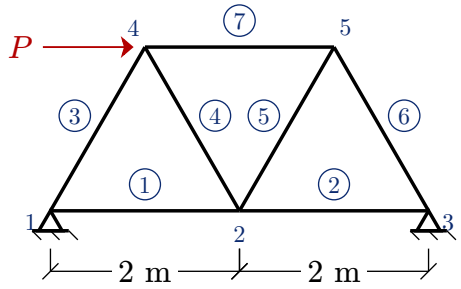


Figura 1: Treliça plana do problema 1.

Como a treliça é bidimensional, cada nó possui dois graus de liberdade, deslocamentos na direção x e y. Para todas as barras, sua matriz de rigidez, portanto, é representada por:

$$K^e = T^T \cdot \bar{K}^e \cdot T$$

Em que T é a matriz de transformação, representada por:

$$T = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 & 0 \\ 0 & 0 & \cos(\theta) & \sin(\theta) \end{bmatrix}$$

A matriz de rigidez nas coordenadas locais é dada por:

$$\bar{K} = \frac{EA}{L} \cdot \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Como todas as barras possuem iguais comprimentos, materiais e seções transversais:

$$\frac{EA}{L} = \frac{180 \cdot 1000 \cdot 0.005}{2}$$

$$\frac{EA}{L} = 450 \text{ MN/m}$$

- Barras 1, 2 e 7

$$T = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 & 0 \\ 0 & 0 & \cos(\theta) & \sin(\theta) \end{bmatrix}$$

$$\theta = 0$$

$$T = \begin{bmatrix} \cos(0) & \sin(0) & 0 & 0 \\ 0 & 0 & \cos(0) & \sin(0) \end{bmatrix}$$

$$T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\bar{K} = 450 \cdot \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$K = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}^T \cdot \begin{bmatrix} 450 & -450 \\ -450 & 450 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$K^{1,2,7} = \begin{bmatrix} 450 & 0 & -450 & 0 \\ 0 & 0 & 0 & 0 \\ -450 & 0 & 450 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- Barras 3 e 5

$$\theta = +60°$$

$$T = \begin{bmatrix} \cos(60°) & \sin(60°) & 0 & 0 \\ 0 & 0 & \cos(60°) & \sin(60°) \end{bmatrix}$$

$$T = \begin{bmatrix} 0.5 & \frac{\sqrt{3}}{2} & 0 & 0 \\ 0 & 0 & 0.5 & \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$K = \begin{bmatrix} 0.5 & \frac{\sqrt{3}}{2} & 0 & 0 \\ 0 & 0 & 0.5 & \frac{\sqrt{3}}{2} \end{bmatrix} \cdot \begin{bmatrix} 450 & -450 \\ -450 & 450 \end{bmatrix} \cdot \begin{bmatrix} 0.5 & \frac{\sqrt{3}}{2} & 0 & 0 \\ 0 & 0 & 0.5 & \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$K^{3,5} = \begin{bmatrix} 112.5 & 194.86 & -112.5 & -194.86 \\ 194.86 & 337.5 & -194.86 & -337.5 \\ -112.5 & -194.86 & 112.5 & 194.86 \\ -194.86 & -337.5 & 194.86 & 337.5 \end{bmatrix}$$

- Barras 4 e 6

$$\theta = +120°$$

$$T = \begin{bmatrix} \cos(120°) & \sin(120°) & 0 & 0 \\ 0 & 0 & \cos(120°) & \sin(120°) \end{bmatrix}$$

$$T = \begin{bmatrix} -0.5 & \frac{\sqrt{3}}{2} & 0 & 0 \\ 0 & 0 & -0.5 & \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$K = \begin{bmatrix} -0.5 & \frac{\sqrt{3}}{2} & 0 & 0 \\ 0 & 0 & -0.5 & \frac{\sqrt{3}}{2} \end{bmatrix} \cdot \begin{bmatrix} 450 & -450 \\ -450 & 450 \end{bmatrix} \cdot \begin{bmatrix} -0.5 & \frac{\sqrt{3}}{2} & 0 & 0 \\ 0 & 0 & -0.5 & \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$K^{4,6} = \begin{bmatrix} 112.5 & -194.86 & -112.5 & 194.86 \\ -194.86 & 337.5 & 194.86 & -337.5 \\ -112.5 & 194.86 & 112.5 & -194.86 \\ 194.86 & -337.5 & -194.86 & 337.5 \end{bmatrix}$$

2.1. Matriz de Rigidez Global

$$K = \begin{bmatrix}
 & u_1^x & u_1^y & u_2^x & u_2^y & u_3^x & u_3^y & u_4^x & u_4^y & u_5^x & u_5^y \\
f_1^x & (450+112.5) & (0+194.86) & (-450) & (0) & () & () & (-112.5) & (-194.86) & () & () \\
f_1^y & (0+194.86) & (337.5) & (0) & (0) & () & () & (-194.86) & (-337.5) & () & () \\
f_2^x & (-450) & (0) & (450+112.5+112.5+450) & (0-194.86+194.86+0) & (-450) & (0) & (-112.5) & (-194.86) & (-112.5) & (-194.86) \\
f_2^y & (0) & (0) & (0-194.86+194.86+0) & (0+337.5+337.5+0) & (0) & (0) & (-194.86) & (-337.5) & (-194.86) & (337.5) \\
f_3^x & () & () & (-450) & (0) & (450+112.5) & (0-194.86) & () & () & (-112.5) & (194.86) \\
f_3^y & () & () & (0) & (0) & (0-194.86) & (0+337.5) & () & () & (194.86) & (-337.5) \\
f_4^x & (-112.5) & (-194.86) & (-112.5) & (-194.86) & () & () & (112.5+450+112.5) & (194.86+0-194.86) & (-450) & (0) \\
f_4^y & (-194.86) & (-337.5) & (-194.86) & (-337.5) & () & () & (194.86+0-194.86) & (337.5+0+337.5) & (0) & (0) \\
f_5^x & () & () & (-112.5) & (-194.86) & (-112.5) & (194.86) & (-450) & (0) & (450+112.5+112.5) & (0+194.86-194.86) \\
f_5^y & () & () & (-194.86) & (337.5) & (194.86) & (-337.5) & (0) & (0) & (0+194.86-194.86) & (0+337.5+337.5)
\end{bmatrix}$$

Somando os elementos de cada posição:

$$K = \begin{bmatrix}
562.5 & 194.86 & -450 & 0 & 0 & 0 & -112.5 & -194.86 & 0 & 0 \\
194.86 & 337.5 & 0 & 0 & 0 & 0 & -194.86 & -337.5 & 0 & 0 \\
-450 & 0 & 1125 & 0 & -450 & 0 & -112.5 & -194.86 & -112.5 & -194.86 \\
0 & 0 & 0 & 675 & 0 & 0 & -194.86 & -337.5 & -194.86 & 337.5 \\
0 & 0 & -450 & 0 & 562.5 & -194.86 & 0 & 0 & -112.5 & 194.86 \\
0 & 0 & 0 & 0 & -194.86 & 337.5 & 0 & 0 & 194.86 & -337.5 \\
-112.5 & -194.86 & -112.5 & -194.86 & 0 & 0 & 675 & 0 & -450 & 0 \\
-194.86 & -337.5 & -194.86 & -337.5 & 0 & 0 & 0 & 675 & 0 & 0 \\
0 & 0 & -112.5 & -194.86 & -112.5 & 194.86 & -450 & 0 & 675 & 0 \\
0 & 0 & -194.86 & 337.5 & 194.86 & -337.5 & 0 & 0 & 0 & 675
\end{bmatrix}$$

2.2. Vetores Força e Deslocamentos

O vetor de deslocamentos u é igual a:

$$u = \begin{bmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ u_{3x} \\ u_{3y} \\ u_{4x} \\ u_{4y} \\ u_{5x} \\ u_{5y} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ u_{2x} \\ u_{2y} \\ 0 \\ 0 \\ u_{4x} \\ u_{4y} \\ u_{5x} \\ u_{5y} \end{bmatrix}$$

Enquanto o vetor de forças é:

$$f = \begin{bmatrix} f_{1x} \\ f_{1y} \\ f_{2x} \\ f_{2y} \\ f_{3x} \\ f_{3y} \\ f_{4x} \\ f_{4y} \\ f_{5x} \\ f_{5y} \end{bmatrix} = \begin{bmatrix} f_{1x} \\ f_{1y} \\ 0 \\ 0 \\ f_{3x} \\ f_{3y} \\ 10 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

2.3. Sistema Linear

Basta resolver então:

$$K \cdot U = F$$

$$K = \begin{bmatrix} 562.5 & 194.86 & -450 & 0 & 0 & 0 & -112.5 & -194.86 & 0 & 0 \\ 194.86 & 337.5 & 0 & 0 & 0 & 0 & -194.86 & -337.5 & 0 & 0 \\ -450 & 0 & 1125 & 0 & -450 & 0 & -112.5 & -194.86 & -112.5 & -194.86 \\ 0 & 0 & 0 & 675 & 0 & 0 & -194.86 & -337.5 & -194.86 & 337.5 \\ 0 & 0 & -450 & 0 & 562.5 & -194.86 & 0 & 0 & -112.5 & 194.86 \\ 0 & 0 & 0 & 0 & -194.86 & 337.5 & 0 & 0 & 194.86 & -337.5 \\ -112.5 & -194.86 & -112.5 & -194.86 & 0 & 0 & 675 & 0 & -450 & 0 \\ -194.86 & -337.5 & -194.86 & -337.5 & 0 & 0 & 0 & 675 & 0 & 0 \\ 0 & 0 & -112.5 & -194.86 & -112.5 & 194.86 & -450 & 0 & 675 & 0 \\ 0 & 0 & -194.86 & 337.5 & 194.86 & -337.5 & 0 & 0 & 0 & 675 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ u_{2x} \\ u_{2y} \\ 0 \\ 0 \\ u_{4x} \\ u_{4y} \\ u_{5x} \\ u_{5y} \end{bmatrix} = \begin{bmatrix} f_{1x} \\ f_{1y} \\ 0 \\ 0 \\ f_{3x} \\ f_{3y} \\ 10 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Utilizando o método da eliminação, temos que:

$$K = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1125 & 0 & 0 & 0 & -112.5 & -194.86 & -112.5 & -194.86 \\ 0 & 0 & 0 & 675 & 0 & 0 & -194.86 & -337.5 & -194.86 & 337.5 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -112.5 & -194.86 & 0 & 0 & 675 & 0 & -450 & 0 \\ 0 & 0 & -194.86 & -337.5 & 0 & 0 & 0 & 675 & 0 & 0 \\ 0 & 0 & -112.5 & -194.86 & 0 & 0 & -450 & 0 & 675 & 0 \\ 0 & 0 & -194.86 & 337.5 & 0 & 0 & 0 & 0 & 0 & 675 \end{bmatrix} \cdot \begin{bmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ u_{3x} \\ u_{3y} \\ u_{4x} \\ u_{4y} \\ u_{5x} \\ u_{5y} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 10 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Portanto, os deslocamentos nodais são iguais a:

$$U = \begin{bmatrix} 0 \\ 0 \\ 5.556 \\ -6.415 \\ 0 \\ 0 \\ 30.556 \\ -4.811 \\ 19.444 \\ -1.604 \end{bmatrix} \cdot 10^{-3} \text{ mm}$$

2.4. Esforço Normal nas Barras

Para cada elemento, calcula-se então os esforços normais em suas coordenadas locais:

$$N = \frac{EA}{L} \cdot T \cdot (u_2 - u_1)$$

- Barra 1:

$$U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 5.556 \\ -6.415 \end{bmatrix}$$

$$U = \begin{bmatrix} 0 \\ 5.556 \end{bmatrix} \cdot 10^{-6} \text{ m}$$

$$N = 450 \cdot 10^3 \cdot (5.556 - 0) \cdot 10^{-6}$$

$$N = +2.5 \text{ kN}$$

Sabendo que as barras 1, 2 e 7 são horizontais, já se calculam os valores diretamente, sem a utilização da matriz de transformação.

- Barra 2:

$$N = 450 \cdot 10^3 \cdot (0 - 5.556) \cdot 10^{-6}$$

$$N = -2.5 \text{ kN}$$

- Barra 7:

$$N = 450 \cdot 10^3 \cdot (19.444 - 30.556) \cdot 10^{-6}$$

$$N = -5.0 \text{ kN}$$

- Barra 3:

$$U = \begin{bmatrix} 0.5 & \frac{\sqrt{3}}{2} & 0 & 0 \\ 0 & 0 & 0.5 & \frac{\sqrt{3}}{2} \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 30.556 \\ -4.811 \end{bmatrix}$$

$$U = \begin{bmatrix} 0 \\ 11.111 \end{bmatrix} \cdot 10^{-6} \text{ m}$$

$$N = 450 \cdot 10^3 \cdot (11.111 - 0) \cdot 10^{-6}$$

$$N = 5 \text{ kN}$$

- Barra 5:

$$U = \begin{bmatrix} 0.5 & \frac{\sqrt{3}}{2} & 0 & 0 \\ 0 & 0 & 0.5 & \frac{\sqrt{3}}{2} \end{bmatrix} \cdot \begin{bmatrix} 5.556 \\ -6.415 \\ 19.444 \\ -1.604 \end{bmatrix}$$

$$U = \begin{bmatrix} -2.778 \\ 8.333 \end{bmatrix} \cdot 10^{-6} \text{ m}$$

$$N = 450 \cdot 10^3 \cdot (8.333 + 2.778) \cdot 10^{-6}$$

$$N = 5 \text{ kN}$$

- Barra 4:

$$U = \begin{bmatrix} -0.5 & \frac{\sqrt{3}}{2} & 0 & 0 \\ 0 & 0 & -0.5 & \frac{\sqrt{3}}{2} \end{bmatrix} \cdot \begin{bmatrix} 5.556 \\ -6.415 \\ 30.556 \\ -4.811 \end{bmatrix}$$

$$U = \begin{bmatrix} -8.333 \\ -19.444 \end{bmatrix} \cdot 10^{-6} \text{ m}$$

$$N = 450 \cdot 10^3 \cdot (-19.444 + 8.333) \cdot 10^{-6}$$

$$N = -5 \text{ kN}$$

- Barra 6:

$$U = \begin{bmatrix} -0.5 & \frac{\sqrt{3}}{2} & 0 & 0 \\ 0 & 0 & -0.5 & \frac{\sqrt{3}}{2} \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 19.444 \\ -1.604 \end{bmatrix}$$

$$U = \begin{bmatrix} 0 \\ -11.111 \end{bmatrix} \cdot 10^{-6} \text{ m}$$

$$N = 450 \cdot 10^3 \cdot (-11.111 - 0) \cdot 10^{-6}$$

$$N = -5 \text{ kN}$$

2.5. Tabela de Esforços Axiais

Barra	N (kN)	Natureza
1	2.5	Tração
2	2.5	Compressão
3	5.0	Tração
4	5.0	Compressão
5	5.0	Tração
6	5.0	Compressão
7	5.0	Compressão

Tabela 2: Esforços axiais nas barras da treliça.

3. Exercício II

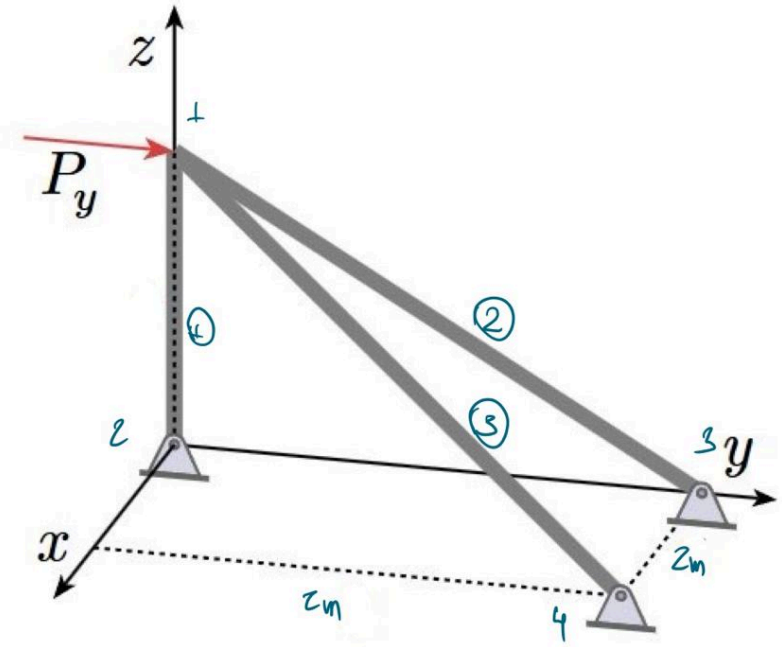


Figura 2: Treliça espacial do problema 2.

3.1. Matrizes de rigidez

Sabendo que a matriz de transformação T para um elemento de treliça espacial é:

$$T = \begin{bmatrix} l & m & n & 0 & 0 & 0 \\ 0 & 0 & 0 & l & m & n \end{bmatrix}$$

que pode ser obtida a partir do vetor unitário que representa a direção da barra:

$$\hat{n} = \frac{\vec{B}}{\left\|\vec{B}\right\|} = \begin{pmatrix} l \\ m \\ n \end{pmatrix}$$

Assim:

$$K = \frac{EA}{L} \begin{bmatrix} \Lambda & -\Lambda \\ -\Lambda & \Lambda \end{bmatrix}$$

Em que:

$$\Lambda = \begin{bmatrix} l^2 & lm & ln \\ lm & m^2 & mn \\ ln & mn & n^2 \end{bmatrix}$$

- Barra 1:

$$B_1 = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$$

$$L_1 = 3 \text{ m}$$

$$\widehat{n_1} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\Lambda_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\frac{E \cdot A}{L_1} = \frac{180 \cdot 10^3 \cdot 0.005}{3}$$

$$\frac{E \cdot A}{L_1} = 300 \text{ MN/m}$$

$$K = 300 \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 \end{bmatrix}$$

$$K_1 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 300 & 0 & 0 & -300 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -300 & 0 & 0 & 300 \end{bmatrix}$$

- Barra 2:

$$B_2 = \begin{pmatrix} 0-0 \\ 2-0 \\ 0-3 \end{pmatrix}$$

$$B_2 = \begin{pmatrix} 0 \\ 2 \\ -3 \end{pmatrix}$$

$$L_2 = \sqrt{2^2 + 3^2}$$

$$L_2 = 3.606 \text{ m}$$

$$\widehat{n_2} = \frac{\begin{pmatrix} 0 \\ 2 \\ -3 \end{pmatrix}}{3.606}$$

$$\widehat{n_2} = \begin{pmatrix} 0 \\ 0.555 \\ -0.832 \end{pmatrix}$$

$$\Lambda_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0.555^2 & 0.555 \cdot (-0.832) \\ 0 & 0.555 \cdot (-0.832) & (-0.832)^2 \end{bmatrix}$$

$$\Lambda_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0.308 & -0.461 \\ 0 & -0.461 & 0.691 \end{bmatrix}$$

$$\frac{E \cdot A}{L_2} = \frac{180 \cdot 10^3 \cdot 0.005}{3.606}$$

$$\frac{E \cdot A}{L_2} = 249.584 \text{ MN/m}$$

$$K_2 = 249.584 \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.308 & -0.461 & 0 & -0.308 & 0.461 \\ 0 & -0.461 & 0.691 & 0 & 0.461 & -0.691 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -0.308 & 0.461 & 0 & 0.308 & -0.461 \\ 0 & 0.461 & -0.691 & 0 & -0.461 & 0.691 \end{bmatrix}$$

$$K_2 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 76.868 & -115.109 & 0 & -76.868 & 115.109 \\ 0 & -115.109 & 172.353 & 0 & 115.109 & -172.353 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -76.868 & 115.109 & 0 & 76.868 & -115.109 \\ 0 & 115.109 & -172.353 & 0 & -115.109 & 172.353 \end{bmatrix}$$

- Barra 3:

$$L_3 = \left\| \overrightarrow{B_3} \right\|$$

$$\overrightarrow{B_3} = \begin{pmatrix} 2-0 \\ 2-0 \\ 0-3 \end{pmatrix}$$

$$\overrightarrow{B_3} = \begin{pmatrix} 2 \\ 2 \\ -3 \end{pmatrix}$$

$$L_3 = \sqrt{2^2 + 2^2 + 3^2}$$

$$L_3 = 4.123 \text{ m}$$

$$\widehat{n_3} = \frac{\begin{pmatrix} 2 \\ 2 \\ -3 \end{pmatrix}}{4.123}$$

$$\widehat{n_3} = \begin{pmatrix} 0.485 \\ 0.485 \\ -0.728 \end{pmatrix}$$

$$\Lambda_3 = \begin{bmatrix} 0.485^2 & 0.485^2 & 0.485 \cdot (-0.728) \\ 0.485^2 & 0.485^2 & 0.485 \cdot (-0.728) \\ 0.485 \cdot (-0.728) & 0.485 \cdot (-0.728) & (-0.728)^2 \end{bmatrix}$$

$$\Lambda_3 = \begin{bmatrix} 0.235 & 0.235 & -0.353 \\ 0.235 & 0.235 & -0.353 \\ -0.353 & -0.353 & 0.529 \end{bmatrix}$$

$$\frac{E \cdot A}{L_3} = \frac{180 \cdot 10^3 \cdot 0.005}{4.123}$$

$$\frac{E \cdot A}{L_3} = 218.288 \text{ MN/m}$$

$$K_3 = 218.288 \begin{bmatrix} 0.235 & 0.235 & -0.353 & -0.235 & -0.235 & 0.353 \\ 0.235 & 0.235 & -0.353 & -0.235 & -0.235 & 0.353 \\ -0.353 & -0.353 & 0.529 & 0.353 & 0.353 & -0.529 \\ -0.235 & -0.235 & 0.353 & 0.235 & 0.235 & -0.353 \\ -0.235 & -0.235 & 0.353 & 0.235 & 0.235 & -0.353 \\ 0.353 & 0.353 & -0.529 & -0.353 & -0.353 & 0.529 \end{bmatrix}$$

$$K_3 = \begin{bmatrix} 51.365 & 51.365 & -77.047 & -51.365 & -51.365 & 77.047 \\ 51.365 & 51.365 & -77.047 & -51.365 & -51.365 & 77.047 \\ -77.047 & -77.047 & 115.57 & 77.047 & 77.047 & -115.57 \\ -51.365 & -51.365 & 77.047 & 51.365 & 51.365 & -77.047 \\ -51.365 & -51.365 & 77.047 & 51.365 & 51.365 & -77.047 \\ 77.047 & 77.047 & -115.57 & -77.047 & -77.047 & 115.57 \end{bmatrix}$$

3.2. Matriz de Rigidez Global

$$K = \begin{bmatrix}
 & u_{1x} & u_{1y} & u_{1z} & u_{2x} & u_{2y} & u_{2z} & u_{3x} & u_{3y} & u_{3z} & u_{4x} & u_{4y} & u_{4z} \\
f_{1x} & (0+0+51.365) & (0+0+51.365) & (0+0-77.047) & (0) & (0) & (0) & (0) & (0) & (0) & (-51.365) & (-51.365) & (77.047) \\
f_{1y} & (0+0+51.365) & (0+76.868+51.365) & (0-115.109-77.047) & (0) & (0) & (0) & (0) & (-76.868) & (115.109) & (-51.365) & (-51.365) & (77.047) \\
f_{1z} & (0+0-77.047) & (0-115.109-77.047) & (300+172.353+115.57) & (0) & (0) & (-300) & (0) & (115.109) & (-172.353) & (77.047) & (77.047) & (-115.57) \\
f_{2x} & (0) & (0) & (0) & (0) & (0) & (0) & () & () & () & () & () & () \\
f_{2y} & (0) & (0) & (0) & (0) & (0) & (0) & () & () & () & () & () & () \\
f_{2z} & (0) & (0) & (-300) & (0) & (0) & (300) & () & () & () & () & () & () \\
f_{3x} & (0) & (0) & (0) & () & () & () & (0) & (0) & (0) & () & () & () \\
f_{3y} & (0) & (-76.868) & (115.109) & () & () & () & (0) & (76.868) & (-115.109) & () & () & () \\
f_{3z} & (0) & (115.109) & (-172.353) & () & () & () & (0) & (-115.109) & (172.353) & () & () & () \\
f_{4x} & (-51.365) & (-51.365) & (77.047) & () & () & () & () & () & () & (51.365) & (51.365) & (-77.047) \\
f_{4y} & (-51.365) & (-51.365) & (77.047) & () & () & () & () & () & () & (51.365) & (51.365) & (-77.047) \\
f_{4z} & (77.047) & (77.047) & (-115.57) & () & () & () & () & () & () & (-77.047) & (-77.047) & (115.57)
\end{bmatrix}$$

$$K = \begin{bmatrix}
51.365 & 51.365 & -77.047 & 0 & 0 & 0 & 0 & 0 & 0 & -51.365 & -51.365 & 77.047 \\
51.365 & 128.233 & -192.156 & 0 & 0 & 0 & 0 & -76.868 & 115.109 & -51.365 & -51.365 & 77.047 \\
-77.047 & -192.156 & 587.923 & 0 & 0 & -300 & 0 & 115.109 & -172.353 & 77.047 & 77.047 & -115.57 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -300 & 0 & 0 & 300 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -76.868 & 115.109 & 0 & 0 & 0 & 0 & 76.868 & -115.109 & 0 & 0 & 0 \\
0 & 115.109 & -172.353 & 0 & 0 & 0 & 0 & -115.109 & 172.353 & 0 & 0 & 0 \\
-51.365 & -51.365 & 77.047 & 0 & 0 & 0 & 0 & 0 & 0 & 51.365 & 51.365 & -77.047 \\
-51.365 & -51.365 & 77.047 & 0 & 0 & 0 & 0 & 0 & 0 & 51.365 & 51.365 & -77.047 \\
77.047 & 77.047 & -115.57 & 0 & 0 & 0 & 0 & 0 & 0 & -77.047 & -77.047 & 115.57
\end{bmatrix}$$

3.3. Vetores Força e Deslocamentos

$$F = \begin{pmatrix} 0 \\ 10 \\ 0 \\ f_{2x} \\ f_{2y} \\ f_{2z} \\ f_{3x} \\ f_{3y} \\ f_{3z} \\ f_{4x} \\ f_{4y} \\ f_{4z} \end{pmatrix}$$

$$U = \begin{pmatrix} u_{1x} \\ u_{1y} \\ u_{1z} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$KU = F$$

$$\begin{bmatrix} 51.365 & 51.365 & -77.047 & 0 & 0 & 0 & 0 & 0 & 0 & -51.365 & -51.365 & 77.047 \\ 51.365 & 128.233 & -192.156 & 0 & 0 & 0 & 0 & -76.868 & 115.109 & -51.365 & -51.365 & 77.047 \\ -77.047 & -192.156 & 587.923 & 0 & 0 & -300 & 0 & 115.109 & -172.353 & 77.047 & 77.047 & -115.57 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -300 & 0 & 0 & 300 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -76.868 & 115.109 & 0 & 0 & 0 & 0 & 76.868 & -115.109 & 0 & 0 & 0 \\ 0 & 115.109 & -172.353 & 0 & 0 & 0 & 0 & -115.109 & 172.353 & 0 & 0 & 0 \\ -51.365 & -51.365 & 77.047 & 0 & 0 & 0 & 0 & 0 & 0 & 51.365 & 51.365 & -77.047 \\ -51.365 & -51.365 & 77.047 & 0 & 0 & 0 & 0 & 0 & 0 & 51.365 & 51.365 & -77.047 \\ 77.047 & 77.047 & -115.57 & 0 & 0 & 0 & 0 & 0 & 0 & -77.047 & -77.047 & 115.57 \end{bmatrix} \begin{pmatrix} u_{1x} \\ u_{1y} \\ u_{1z} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 10 \\ 0 \\ f_{2x} \\ f_{2y} \\ f_{2z} \\ f_{3x} \\ f_{3y} \\ f_{3z} \\ f_{4x} \\ f_{4y} \\ f_{4z} \end{pmatrix}$$

3.4. Resolução do Sistema Linear

$$\begin{bmatrix} 51.365 & 51.365 & -77.047 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 51.365 & 128.233 & -192.156 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -77.047 & -192.156 & 587.923 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} u_{1x} \\ u_{1y} \\ u_{1z} \\ u_{2x} \\ u_{2y} \\ u_{2z} \\ u_{3x} \\ u_{3y} \\ u_{3z} \\ u_{4x} \\ u_{4y} \\ u_{4z} \end{pmatrix} = \begin{pmatrix} 0 \\ 10 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$U = \begin{pmatrix} -0.13 \\ 0.205 \\ 0.05 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \text{ mm}$$

3.5. Esforços Normais

$$N = \frac{E \cdot A}{L} \cdot (u_2 - u_1)$$

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{bmatrix} l & m & n & 0 & 0 & 0 \\ 0 & 0 & 0 & l & m & n \end{bmatrix} \cdot \begin{pmatrix} u_{1x} \\ u_{1y} \\ u_{1z} \\ u_{2x} \\ u_{2y} \\ u_{2z} \end{pmatrix}$$

- Barra 1:

$$u = \begin{pmatrix} 0 \\ 0.05 \end{pmatrix} \text{ mm}$$

Barra na direção z, portanto:

$$N = 300 \cdot 10^3 \cdot 0.05 \cdot 10^{-3}$$

$$N = +15 \text{ kN}$$

- Barra 2:

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{bmatrix} 0 & 0.555 & -0.832 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.555 & -0.832 \end{bmatrix} \cdot \begin{pmatrix} -0.13 \\ 0.205 \\ 0.05 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0.072 \\ 0 \end{pmatrix}$$

$$N = 249.584 \cdot 10^3 \cdot (-0.072) \cdot 10^{-3}$$

$$N = -18.014 \text{ kN}$$

- Barra 3:

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{bmatrix} 0.485 & 0.485 & -0.728 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.485 & 0.485 & -0.728 \end{bmatrix} \cdot \begin{pmatrix} -0.13 \\ 0.205 \\ 0.05 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$N = 0$$

3.6. Tabela de Deslocamentos e Forças

Nó	GDL	Deslocamento (mm)	Força (kN)
1	u_x	-0.13	0
1	u_y	0.205	10
1	u_z	0.05	0
2	u_x	0	f_{2x}
2	u_y	0	f_{2y}
2	u_z	0	f_{2z}
3	u_x	0	f_{3x}
3	u_y	0	f_{3y}
3	u_z	0	f_{3z}
4	u_x	0	f_{4x}
4	u_y	0	f_{4y}
4	u_z	0	f_{4z}

Tabela 3: Deslocamentos e forças nodais.

3.7. Tabela de Esforços Axiais

Barra	N (kN)	Natureza
1	15	Tração
2	18.014	Compressão
3	0	Nula

Tabela 4: Esforços axiais nas barras da treliça.

4. Exercício III

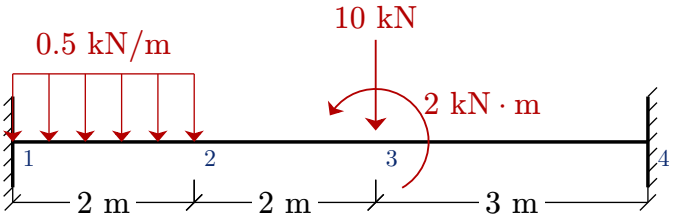


Figura 3: Viga biengastada do problema 3.

4.1. Solução Analítica

4.1.1. Equações utilizando funções de Macaulay

$$Q(x) = V_a - 0.5x + 0.5\langle x-2\rangle - 10\langle x-4\rangle^0 + V_d\langle x-7\rangle^0$$

$$\int Q(x)\,\mathrm{d}x = M(x) + M(0)$$

$$M(0) = 0$$

$$M(x) = -M_a + V_a x - \frac{0.5x^2}{2!} + \frac{0.5\langle x-2\rangle^2}{2!} - 10\langle x-4\rangle - 2\langle x-4\rangle^0 + M_d\langle x-7\rangle^0$$

$$\frac{\int M(x)\,\mathrm{d}x}{EI} = \Theta(x) + \Theta(0)$$

$$\Theta(0) = 0$$

$$\Theta(x)\cdot EI = -M_a x + \frac{V_a x^2}{2!} - \frac{0.5x^3}{3!} + \frac{0.5\langle x-2\rangle^3}{3!} - \frac{10\langle x-4\rangle^2}{2!} - 2\langle x-4\rangle$$

$$\int \theta(x)\,\mathrm{d}x = v(x) + v(0)$$

$$v(x)\cdot EI = -\frac{M_a x^2}{2!} + \frac{V_a x^3}{3!} - \frac{0.5x^4}{4!} + \frac{0.5\langle x-2\rangle^4}{4!} - \frac{10\langle x-4\rangle^3}{3!} - \frac{2\langle x-4\rangle^2}{2!}$$

4.1.2. Aplicando condições de contorno

$$\Theta(7)\cdot EI = 0 \Rightarrow 0 = -7M_a + \frac{V_a\cdot 49}{2} - \frac{0.5\cdot 343}{6} + \frac{0.5\cdot 125}{6} - \frac{10\cdot 9}{2} - 2\cdot 3$$

$$0 = -7M_a + 24.5V_a - 28.583 + 10.417 - 45 - 6$$

$$7M_a - 24.5V_a = -69.167$$

$$v(7) \cdot EI = 0 \Rightarrow 0 = -\frac{M_a \cdot 49}{2} + \frac{V_a \cdot 343}{6} - \frac{0.5 \cdot 2401}{24} + \frac{0.5 \cdot 625}{24} - \frac{10 \cdot 27}{6} - \frac{2 \cdot 9}{2}$$

$$0 = -24.5M_a + 57.167V_a - 50.021 + 13.021 - 45 - 9$$

$$24.5M_a - \left(\frac{343}{6}\right)V_a = -91$$

4.1.3. Reações de apoio

$$\begin{cases} 7M_a - 24.5V_a = -69.167 \\ 24.5M_a - \left(\frac{343}{6}\right)V_a = -91 \end{cases}$$

$$V_a = 5.286 \ \text{kN}$$

$$M_a = 8.619 \ \text{kN} \cdot \text{m}$$

$$\sum F_y = 0$$

$$0 = V_a + V_d - 10 - 0.5 \cdot 2$$

$$V_d = 10 + 1 - 5.286$$

$$V_d = 5.714 \ \text{kN}$$

$$\sum M_{\text{em A}} = 0$$

$$0 = M_a - \frac{2^2 \cdot q}{2} + M - P \cdot 4 + M_d + 7 \cdot V_d$$

$$0 = 8.619 - 1 + 2 - 40 + M_d + 7 \cdot 5.714$$

$$M_d = -9.617 \ \text{kN} \cdot \text{m}$$

4.1.4. Deslocamentos

$$v(x) \cdot 180 \cdot 10^6 \cdot 2.1 \cdot 10^{-6} = -\frac{8.619 \cdot x^2}{2!} + \frac{5.286x^3}{3!} - \frac{0.5x^4}{4!} + \frac{0.5\langle x-2\rangle^4}{4!} - \frac{10\langle x-4\rangle^3}{3!} - \frac{2\langle x-4\rangle^2}{2!}$$

$$v(x) \cdot 378 = -4.310x^2 + 0.881x^3 - 0.0208x^4 + 0.0208\langle x-2\rangle^4 - 1.667\langle x-4\rangle^3 - \langle x-4\rangle^2$$

$$v(2) = -\frac{10.524}{378}$$

$$v(2) = -27.841 \ \text{mm}$$

$$v(4) = -\frac{17.571}{378}$$

$$v(4) = -46.484 \text{ mm}$$

4.2. Rigidez Direta

A matriz de rigidez de uma viga de Euler-Bernoulli possui diferentes equações a depender do seu carregamento. Para um elemento com uma carga q distribuída uniformemente, ela é expressa por:

$$\frac{EI}{l^3}\begin{pmatrix} 12 & 6l & -12 & 6l \\ 6l & 4l^2 & -6l & 2l^2 \\ -12 & -6l & 12 & -6l \\ 6l & 2l^2 & -6l & 4l^2 \end{pmatrix}\begin{pmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} V_1 \\ M_1 \\ V_2 \\ M_2 \end{pmatrix} + \frac{ql}{12}\begin{pmatrix} 6 \\ l \\ 6 \\ -l \end{pmatrix}$$

- Elementos 1 e 2

$$K = \frac{180 \cdot 10^6 \cdot 2.1 \cdot 10^{-6}}{2^3}\begin{pmatrix} 12 & 6 \cdot 2 & -12 & 6 \cdot 2 \\ 6 \cdot 2 & 4 \cdot 2^2 & -6 \cdot 2 & 2 \cdot 2^2 \\ -12 & -6 \cdot 2 & 12 & -6 \cdot 2 \\ 6 \cdot 2 & 2 \cdot 2^2 & -6 \cdot 2 & 4 \cdot 2^2 \end{pmatrix}$$

$$K = 47.25\begin{pmatrix} 12 & 12 & -12 & 12 \\ 12 & 16 & -12 & 8 \\ -12 & -12 & 12 & -12 \\ 12 & 8 & -12 & 16 \end{pmatrix}$$

$$K = \begin{pmatrix} 567 & 567 & -567 & 567 \\ 567 & 756 & -567 & 378 \\ -567 & -567 & 567 & -567 \\ 567 & 378 & -567 & 756 \end{pmatrix}$$

- Elemento 3

$$K = \frac{180 \cdot 10^6 \cdot 2.1 \cdot 10^{-6}}{3^3}\begin{pmatrix} 12 & 6 \cdot 3 & -12 & 6 \cdot 3 \\ 6 \cdot 3 & 4 \cdot 3^2 & -6 \cdot 3 & 2 \cdot 3^2 \\ -12 & -6 \cdot 3 & 12 & -6 \cdot 3 \\ 6 \cdot 3 & 2 \cdot 3^2 & -6 \cdot 3 & 4 \cdot 3^2 \end{pmatrix}$$

$$K = 14\begin{pmatrix} 12 & 18 & -12 & 18 \\ 18 & 36 & -18 & 18 \\ -12 & -18 & 12 & -18 \\ 18 & 18 & -18 & 36 \end{pmatrix}$$

$$K = \begin{pmatrix} 168 & 252 & -168 & 252 \\ 252 & 504 & -252 & 252 \\ -168 & -252 & 168 & -252 \\ 252 & 252 & -252 & 504 \end{pmatrix}$$

4.2.1. Matriz de Rigidez Global

$$K = \begin{bmatrix} 567 & 567 & -567 & 567 & 0 & 0 & 0 & 0 \\ 567 & 756 & -567 & 378 & 0 & 0 & 0 & 0 \\ -567 & -567 & 567+567 & 567-567 & -567 & 567 & 0 & 0 \\ 567 & 378 & -567+567 & 756+756 & -567 & 378 & 0 & 0 \\ 0 & 0 & -567 & -567 & 567+168 & -567+252 & -168 & 252 \\ 0 & 0 & 567 & 378 & -567+252 & 756+504 & -252 & 252 \\ 0 & 0 & 0 & 0 & -168 & -252 & 168 & -252 \\ 0 & 0 & 0 & 0 & 252 & 252 & -252 & 504 \end{bmatrix}$$

Somando os elementos de cada posição:

$$K = \begin{bmatrix} 567 & 567 & -567 & 567 & 0 & 0 & 0 & 0 \\ 567 & 756 & -567 & 378 & 0 & 0 & 0 & 0 \\ -567 & -567 & 1134 & 0 & -567 & 567 & 0 & 0 \\ 567 & 378 & 0 & 1512 & -567 & 378 & 0 & 0 \\ 0 & 0 & -567 & -567 & 735 & -315 & -168 & 252 \\ 0 & 0 & 567 & 378 & -315 & 1260 & -252 & 252 \\ 0 & 0 & 0 & 0 & -168 & -252 & 168 & -252 \\ 0 & 0 & 0 & 0 & 252 & 252 & -252 & 504 \end{bmatrix}$$

4.2.2. Vetores Força e Deslocamentos

$$F = \begin{pmatrix} V_1 \\ M_1 \\ 0 \\ 0 \\ -10 \\ 2 \\ V_2 \\ M_2 \end{pmatrix} - \frac{0.5 \cdot 2}{12} \begin{pmatrix} 6 \\ 2 \\ 6 \\ -2 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$F = \begin{pmatrix} V_1 \\ M_1 \\ 0 \\ 0 \\ -10 \\ 2 \\ V_2 \\ M_2 \end{pmatrix} - \begin{pmatrix} 0.5 \\ 0.167 \\ 0.5 \\ -0.167 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$F = \begin{pmatrix} V_1 - 0.5 \\ M_1 - 0.167 \\ -0.5 \\ 0.167 \\ -10 \\ 2 \\ V_2 \\ M_2 \end{pmatrix}$$

$$U = \begin{pmatrix} 0 \\ 0 \\ v_2 \\ \theta_2 \\ v_3 \\ \theta_3 \\ 0 \\ 0 \end{pmatrix}$$

4.2.3. Resolução do Sistema Linear

$$KU = F$$

$$\begin{bmatrix} 567 & 567 & -567 & 567 & 0 & 0 & 0 & 0 \\ 567 & 756 & -567 & 378 & 0 & 0 & 0 & 0 \\ -567 & -567 & 1134 & 0 & -567 & 567 & 0 & 0 \\ 567 & 378 & 0 & 1512 & -567 & 378 & 0 & 0 \\ 0 & 0 & -567 & -567 & 735 & -315 & -168 & 252 \\ 0 & 0 & 567 & 378 & -315 & 1260 & -252 & 252 \\ 0 & 0 & 0 & 0 & -168 & -252 & 168 & -252 \\ 0 & 0 & 0 & 0 & 252 & 252 & -252 & 504 \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ v_2 \\ \theta_2 \\ v_3 \\ \theta_3 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} V_1 - 0.5 \\ M_1 - 0.167 \\ -0.5 \\ 0.167 \\ -10 \\ 2 \\ V_2 \\ M_2 \end{pmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1134 & 0 & -567 & 567 & 0 & 0 \\ 0 & 0 & 0 & 1512 & -567 & 378 & 0 & 0 \\ 0 & 0 & -567 & -567 & 735 & -315 & 0 & 0 \\ 0 & 0 & 567 & 378 & -315 & 1260 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \\ v_3 \\ \theta_3 \\ v_4 \\ \theta_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ -0.5 \\ 0.167 \\ -10 \\ 2 \\ 0 \\ 0 \end{pmatrix}$$

$$U = \begin{pmatrix} 0 \\ 0 \\ -27.84 \\ -19.4 \\ -46.485 \\ 8.314 \\ 0 \\ 0 \end{pmatrix} \text{ mm/rad}$$

4.2.4. Reações

$$F = KU$$

$$F' = \begin{bmatrix} 567 & 567 & -567 & 567 & 0 & 0 & 0 & 0 \\ 567 & 756 & -567 & 378 & 0 & 0 & 0 & 0 \\ -567 & -567 & 1134 & 0 & -567 & 567 & 0 & 0 \\ 567 & 378 & 0 & 1512 & -567 & 378 & 0 & 0 \\ 0 & 0 & -567 & -567 & 735 & -315 & -168 & 252 \\ 0 & 0 & 567 & 378 & -315 & 1260 & -252 & 252 \\ 0 & 0 & 0 & 0 & -168 & -252 & 168 & -252 \\ 0 & 0 & 0 & 0 & 252 & 252 & -252 & 504 \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ -27.84 \\ -19.4 \\ -46.485 \\ 8.314 \\ 0 \\ 0 \end{pmatrix}$$

$$F' = \begin{pmatrix} 4.786 \\ 8.452 \\ -0.5 \\ 0.167 \\ -10 \\ 2 \\ 5.714 \\ -9.619 \end{pmatrix}$$

$$F = F' - \begin{pmatrix} -0.5 \\ -0.167 \\ -0.5 \\ 0.167 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$F = \begin{pmatrix} 5.286 \\ 8.619 \\ 0 \\ 0 \\ -10 \\ 2 \\ 5.714 \\ -9.619 \end{pmatrix}$$

4.2.5. Comparação dos Resultados

A tabela a seguir compara os resultados obtidos pela forma analítica (funções de Macaulay) com os obtidos pelo método da rigidez direta.

Grandeza	**Forma Analítica**	**Rigidez Direta**
V_a (kN)	5.286	5.286
M_a (kN · m)	8.619	8.619
V_d (kN)	5.714	5.714
M_d (kN · m)	−9.617	−9.619
$v(x = 2)$ (mm)	−27.841	−27.84
$v(x = 4)$ (mm)	−46.484	−46.485

Tabela 5: Comparação entre a forma analítica e o método da rigidez direta.

4.2.6. Diagramas de Esforços

Utilizando a biblioteca *matplotlib*, em Python, determinou-se os gráficos solicitados:

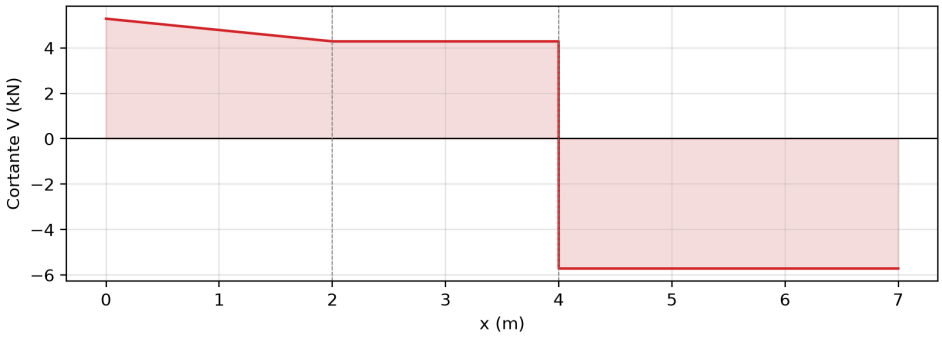


Figura 4: Diagrama de esforço cortante.

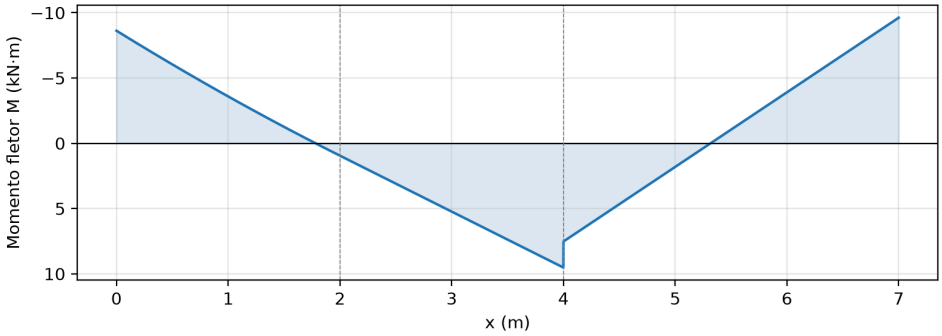


Figura 5: Diagrama de momento fletor.

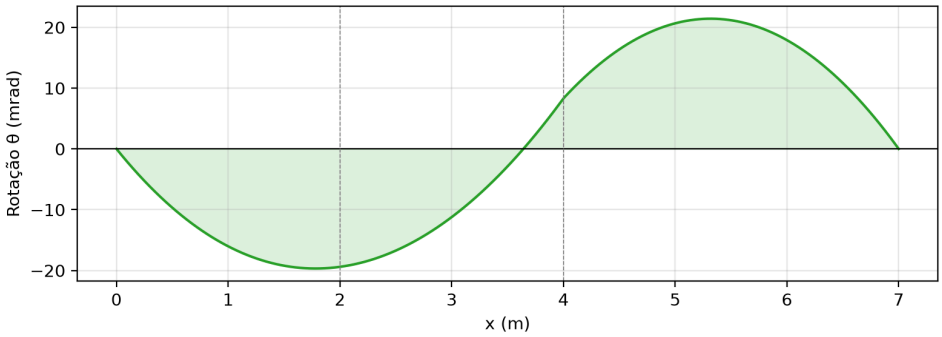


Figura 6: Diagrama de rotação.

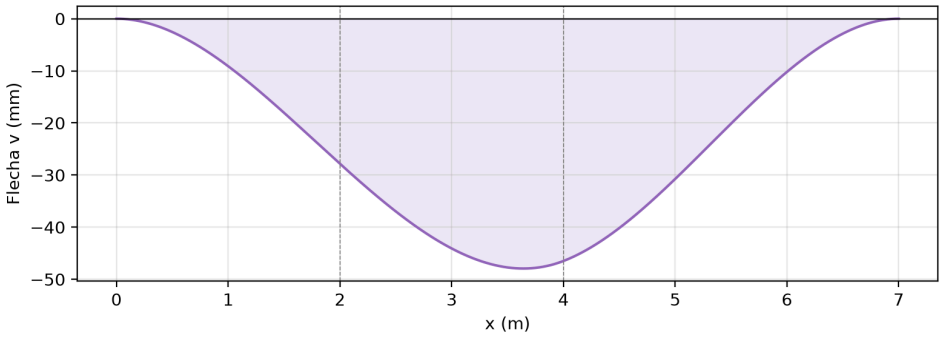
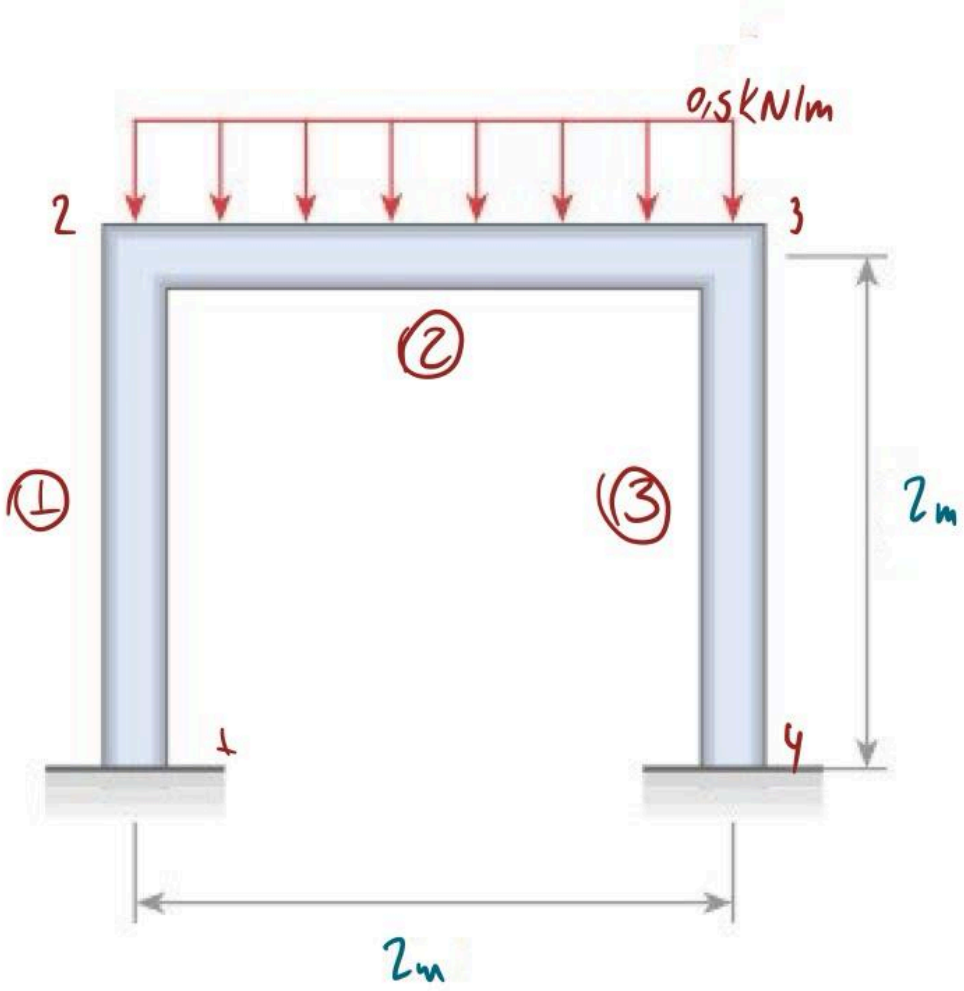


Figura 7: Deslocamento vertical (flecha).

5. Exercício IV



A matriz de rigidez de um pórtico plano constitui-se na combinação entre as matrizes de elementos de viga, nesse caso de Euler-Bernoulli, e de treliça.

$$K = \begin{pmatrix} \frac{EA}{l} & 0 & 0 & -\frac{EA}{l} & 0 & 0 \\ 0 & \frac{12EI}{l^3} & \frac{6EI}{l^2} & 0 & -\frac{12EI}{l^3} & \frac{6EI}{l^2} \\ 0 & \frac{6EI}{l^2} & \frac{4EI}{l} & 0 & -\frac{6EI}{l^2} & \frac{2EI}{l} \\ -\frac{EA}{l} & 0 & 0 & \frac{EA}{l} & 0 & 0 \\ 0 & -\frac{12EI}{l^3} & -\frac{6EI}{l^2} & 0 & \frac{12EI}{l^3} & -\frac{6EI}{l^2} \\ 0 & \frac{6EI}{l^2} & \frac{2EI}{l} & 0 & -\frac{6EI}{l^2} & \frac{4EI}{l} \end{pmatrix}$$

No sistema $KU = F$:

$$\begin{pmatrix} \frac{EA}{l} & 0 & 0 & -\frac{EA}{l} & 0 & 0 \\ 0 & \frac{12EI}{l^3} & \frac{6EI}{l^2} & 0 & -\frac{12EI}{l^3} & \frac{6EI}{l^2} \\ 0 & \frac{6EI}{l^2} & \frac{4EI}{l} & 0 & -\frac{6EI}{l^2} & \frac{2EI}{l} \\ -\frac{EA}{l} & 0 & 0 & \frac{EA}{l} & 0 & 0 \\ 0 & -\frac{12EI}{l^3} & -\frac{6EI}{l^2} & 0 & \frac{12EI}{l^3} & -\frac{6EI}{l^2} \\ 0 & \frac{6EI}{l^2} & \frac{2EI}{l} & 0 & -\frac{6EI}{l^2} & \frac{4EI}{l} \end{pmatrix} \begin{pmatrix} u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} f_1 \\ V_1 \\ M_1 \\ f_2 \\ V_2 \\ M_2 \end{pmatrix}$$

Por fim, como em treliças, é necessária uma matriz de transformação para elementos inclinados:

$$\begin{pmatrix} u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} c & s & 0 & 0 & 0 & 0 \\ -s & c & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & c & s & 0 \\ 0 & 0 & 0 & -s & c & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u_{1x} \\ u_{1y} \\ \theta_1 \\ u_{2x} \\ u_{2y} \\ \theta_2 \end{pmatrix}$$

em que:

$$c = \cos(\theta) \text{ e } s = \sin(\theta)$$

Como todas as barras possuem as mesmas dimensões e rigidez:

$$EA = 180 \cdot 10^6 \cdot 0.005$$

$$EA = 900 \cdot 10^3 \text{ kN}$$

$$EI = 180 \cdot 10^6 \cdot 2.1 \cdot 10^{-6}$$

$$EI = 378 \text{ kN} \cdot \text{m}^2$$

Percebe-se que a rigidez axial desse pórtico é muito maior que a rigidez à flexão, o que implica que as deformações, e a energia de deformação, causadas pela flexão serão muito maiores do que as causadas pelo esforço normal.

Como possuem também as mesmas dimensões, todos possuirão a mesma matriz de rigidez local:

$$\bar{K} = \begin{pmatrix} \frac{900\cdot10^3}{2} & 0 & 0 & -\frac{900\cdot10^3}{2} & 0 & 0 \\ 0 & \frac{12\cdot378}{2^3} & \frac{6\cdot378}{2^2} & 0 & -\frac{12\cdot378}{2^3} & \frac{6\cdot378}{2^2} \\ 0 & \frac{6\cdot378}{2^2} & \frac{4\cdot378}{2} & 0 & -\frac{6\cdot378}{2^2} & \frac{2\cdot378}{2} \\ -\frac{900\cdot10^3}{2} & 0 & 0 & \frac{900\cdot10^3}{2} & 0 & 0 \\ 0 & -\frac{12\cdot378}{2^3} & -\frac{6\cdot378}{2^2} & 0 & \frac{12\cdot378}{2^3} & -\frac{6\cdot378}{2^2} \\ 0 & \frac{6\cdot378}{2^2} & \frac{2\cdot378}{2} & 0 & -\frac{6\cdot378}{2^2} & \frac{4\cdot378}{2} \end{pmatrix}$$

$$\bar{K} = 10^3 \begin{pmatrix} 450 & 0 & 0 & -450 & 0 & 0 \\ 0 & 0.567 & 0.567 & 0 & -0.567 & 0.567 \\ 0 & 0.567 & 0.756 & 0 & -0.567 & 0.378 \\ -450 & 0 & 0 & 450 & 0 & 0 \\ 0 & -0.567 & -0.567 & 0 & 0.567 & -0.567 \\ 0 & 0.567 & 0.378 & 0 & -0.567 & 0.756 \end{pmatrix}$$

- Elementos 1 e 3

$$\theta = 90°$$

$$T_{1,3} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$K = T^T \bar{K} T$$

$$K_{1,3} = 10^3 \begin{pmatrix} 0.567 & 0 & -0.567 & -0.567 & 0 & -0.567 \\ 0 & 450 & 0 & 0 & -450 & 0 \\ -0.567 & 0 & 0.756 & 0.567 & 0 & 0.378 \\ -0.567 & 0 & 0.567 & 0.567 & 0 & 0.567 \\ 0 & -450 & 0 & 0 & 450 & 0 \\ -0.567 & 0 & 0.378 & 0.567 & 0 & 0.756 \end{pmatrix}$$

- Elemento 2

O eixo do elemento 2 coincide com o eixo global escolhido, portanto:

$$\theta = 0$$

Dessa forma, a matriz de rigidez global do elemento 2 é:

$$K_2 = \begin{pmatrix} \frac{900 \cdot 10^3}{2} & 0 & 0 & -\frac{900 \cdot 10^3}{2} & 0 & 0 \\ 0 & \frac{12 \cdot 378}{2^3} & \frac{6 \cdot 378}{2^2} & 0 & -\frac{12 \cdot 378}{2^3} & \frac{6 \cdot 378}{2^2} \\ 0 & \frac{6 \cdot 378}{2^2} & \frac{4 \cdot 378}{2} & 0 & -\frac{6 \cdot 378}{2^2} & \frac{2 \cdot 378}{2} \\ -\frac{900 \cdot 10^3}{2} & 0 & 0 & \frac{900 \cdot 10^3}{2} & 0 & 0 \\ 0 & -\frac{12 \cdot 378}{2^3} & -\frac{6 \cdot 378}{2^2} & 0 & \frac{12 \cdot 378}{2^3} & -\frac{6 \cdot 378}{2^2} \\ 0 & \frac{6 \cdot 378}{2^2} & \frac{2 \cdot 378}{2} & 0 & -\frac{6 \cdot 378}{2^2} & \frac{4 \cdot 378}{2} \end{pmatrix}$$

5.1. Matriz de Rigidez Global

$$K = 10^3 \begin{bmatrix}
 & u_1 & v_1 & \theta_1 & u_2 & v_2 & \theta_2 & u_3 & v_3 & \theta_3 & u_4 & v_4 & \theta_4 \\
f_1 & (0.567) & (0) & (-0.567) & (-0.567) & (0) & (-0.567) & () & () & () & () & () & () \\
V_1 & (0) & (450) & (0) & (0) & (-450) & (0) & () & () & () & () & () & () \\
M_1 & (-0.567) & (0) & (0.756) & (0.567) & (0) & (0.378) & () & () & () & () & () & () \\
f_2 & (-0.567) & (0) & (0.567) & (0.567+450) & (0+0) & (0.567+0) & (-450) & (0) & (0) & () & () & () \\
V_2 & (0) & (-450) & (0) & (0+0) & (450+0.567) & (0+0.567) & (0) & (-0.567) & (0.567) & () & () & () \\
M_2 & (-0.567) & (0) & (0.378) & (0.567+0) & (0+0.567) & (0.756+0.756) & (0) & (-0.567) & (0.378) & () & () & () \\
f_3 & () & () & () & (-450) & (0) & (0) & (450+0.567) & (0+0) & (0-0.567) & (-0.567) & (0) & (-0.567) \\
V_3 & () & () & () & (0) & (-0.567) & (-0.567) & (0+0) & (0.567+450) & (-0.567+0) & (0) & (-450) & (0) \\
M_3 & () & () & () & (0) & (0.567) & (0.378) & (0-0.567) & (-0.567+0) & (0.756+0.756) & (0.567) & (0) & (0.378) \\
f_4 & () & () & () & () & () & () & (-0.567) & (0) & (0.567) & (0.567) & (0) & (0.567) \\
V_4 & () & () & () & () & () & () & (0) & (-450) & (0) & (0) & (450) & (0) \\
M_4 & () & () & () & () & () & () & (-0.567) & (0) & (0.378) & (0.567) & (0) & (0.756)
\end{bmatrix}$$

$$K = 10^3 \begin{bmatrix}
0.567 & 0 & -0.567 & -0.567 & 0 & -0.567 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 450 & 0 & 0 & -450 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-0.567 & 0 & 0.756 & 0.567 & 0 & 0.378 & 0 & 0 & 0 & 0 & 0 & 0 \\
-0.567 & 0 & 0.567 & 450.567 & 0 & 0.567 & -450 & 0 & 0 & 0 & 0 & 0 \\
0 & -450 & 0 & 0 & 450.567 & 0.567 & 0 & -0.567 & 0.567 & 0 & 0 & 0 \\
-0.567 & 0 & 0.378 & 0.567 & 0.567 & 1.512 & 0 & -0.567 & 0.378 & 0 & 0 & 0 \\
0 & 0 & 0 & -450 & 0 & 0 & 450.567 & 0 & -0.567 & -0.567 & 0 & -0.567 \\
0 & 0 & 0 & 0 & -0.567 & -0.567 & 0 & 450.567 & -0.567 & 0 & -450 & 0 \\
0 & 0 & 0 & 0 & 0.567 & 0.378 & -0.567 & -0.567 & 1.512 & 0.567 & 0 & 0.378 \\
0 & 0 & 0 & 0 & 0 & 0 & -0.567 & 0 & 0.567 & 0.567 & 0 & 0.567 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -450 & 0 & 0 & 450 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -0.567 & 0 & 0.378 & 0.567 & 0 & 0.756
\end{bmatrix}$$

5.2. Vetores Força e Deslocamento

$$F = \begin{pmatrix} f_1 \\ V_1 \\ M_1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ f_4 \\ V_4 \\ M_4 \end{pmatrix}$$

$$F' = F + \frac{ql}{12} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 6 \\ l \\ 0 \\ 6 \\ -l \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$F' = \begin{pmatrix} f_1 \\ V_1 \\ M_1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ f_4 \\ V_4 \\ M_4 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0.5 \\ 0.167 \\ 0 \\ 0.5 \\ -0.167 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$F' = \begin{pmatrix} f_1 \\ V_1 \\ M_1 \\ 0 \\ -0.5 \\ -0.167 \\ 0 \\ -0.5 \\ 0.167 \\ f_4 \\ V_4 \\ M_4 \end{pmatrix} \text{ kN}$$

$$U = \begin{pmatrix} u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \\ u_3 \\ v_3 \\ \theta_3 \\ u_4 \\ v_4 \\ \theta_4 \end{pmatrix}$$

$$U = \begin{pmatrix} 0 \\ 0 \\ 0 \\ u_2 \\ v_2 \\ \theta_2 \\ u_3 \\ v_3 \\ \theta_3 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

5.3. Resolução do Sistema Linear

$$KU = F$$

$$10^3 \begin{bmatrix} 0.567 & 0 & -0.567 & -0.567 & 0 & -0.567 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 450 & 0 & 0 & -450 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -0.567 & 0 & 0.756 & 0.567 & 0 & 0.378 & 0 & 0 & 0 & 0 & 0 & 0 \\ -0.567 & 0 & 0.567 & 450.567 & 0 & 0.567 & -450 & 0 & 0 & 0 & 0 & 0 \\ 0 & -450 & 0 & 0 & 450.567 & 0.567 & 0 & -0.567 & 0.567 & 0 & 0 & 0 \\ -0.567 & 0 & 0.378 & 0.567 & 0.567 & 1.512 & 0 & -0.567 & 0.378 & 0 & 0 & 0 \\ 0 & 0 & 0 & -450 & 0 & 0 & 450.567 & 0 & -0.567 & -0.567 & 0 & -0.567 \\ 0 & 0 & 0 & 0 & -0.567 & -0.567 & 0 & 450.567 & -0.567 & 0 & -450 & 0 \\ 0 & 0 & 0 & 0 & 0.567 & 0.378 & -0.567 & -0.567 & 1.512 & 0.567 & 0 & 0.378 \\ 0 & 0 & 0 & 0 & 0 & 0 & -0.567 & 0 & 0.567 & 0.567 & 0 & 0.567 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -450 & 0 & 0 & 450 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -0.567 & 0 & 0.378 & 0.567 & 0 & 0.756 \end{bmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \\ u_2 \\ v_2 \\ \theta_2 \\ u_3 \\ v_3 \\ \theta_3 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} f_1 \\ V_1 \\ M_1 \\ 0 \\ -0.5 \\ -0.167 \\ 0 \\ -0.5 \\ 0.167 \\ f_4 \\ V_4 \\ M_4 \end{pmatrix}$$

$$10^3 \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 450.567 & 0 & 0.567 & -450 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 450.567 & 0.567 & 0 & -0.567 & 0.567 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.567 & 0.567 & 1.512 & 0 & -0.567 & 0.378 & 0 & 0 & 0 \\ 0 & 0 & 0 & -450 & 0 & 0 & 450.567 & 0 & -0.567 & -0.567 & 0 & 0 \\ 0 & 0 & 0 & 0 & -0.567 & -0.567 & 0 & 450.567 & -0.567 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.567 & 0.378 & -0.567 & -0.567 & 1.512 & 0.567 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{pmatrix} u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \\ u_3 \\ v_3 \\ \theta_3 \\ u_4 \\ v_4 \\ \theta_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -0.5 \\ -0.167 \\ 0 \\ -0.5 \\ 0.167 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$U = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0.295 \\ -0.001 \\ -0.295 \\ 0.295 \\ -0.001 \\ 0.295 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

5.4. Determinação das Reações

$$F' = KU$$

$$F' = \begin{bmatrix} 0.567 & 0 & -0.567 & -0.567 & 0 & -0.567 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 450 & 0 & 0 & -450 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -0.567 & 0 & 0.756 & 0.567 & 0 & 0.378 & 0 & 0 & 0 & 0 & 0 & 0 \\ -0.567 & 0 & 0.567 & 450.567 & 0 & 0.567 & -450 & 0 & 0 & 0 & 0 & 0 \\ 0 & -450 & 0 & 0 & 450.567 & 0.567 & 0 & -0.567 & 0.567 & 0 & 0 & 0 \\ -0.567 & 0 & 0.378 & 0.567 & 0.567 & 1.512 & 0 & -0.567 & 0.378 & 0 & 0 & 0 \\ 0 & 0 & 0 & -450 & 0 & 0 & 450.567 & 0 & -0.567 & -0.567 & 0 & -0.567 \\ 0 & 0 & 0 & 0 & -0.567 & -0.567 & 0 & 450.567 & -0.567 & 0 & -450 & 0 \\ 0 & 0 & 0 & 0 & 0.567 & 0.378 & -0.567 & -0.567 & 1.512 & 0.567 & 0 & 0.378 \\ 0 & 0 & 0 & 0 & 0 & 0 & -0.567 & 0 & 0.567 & 0.567 & 0 & 0.567 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -450 & 0 & 0 & 450 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -0.567 & 0 & 0.378 & 0.567 & 0 & 0.756 \end{bmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0.295 \\ -0.001 \\ -0.295 \\ 0.295 \\ -0.001 \\ 0.295 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$F' = \begin{pmatrix} 0 \\ 0.5 \\ 0.056 \\ 0 \\ -0.5 \\ -0.167 \\ 0 \\ -0.5 \\ 0.167 \\ 0 \\ 0.5 \\ -0.056 \end{pmatrix}$$

$$F = \begin{pmatrix} 0 \\ 0.5 \\ 0.056 \\ 0 \\ -0.5 \\ -0.167 \\ 0 \\ -0.5 \\ 0.167 \\ 0 \\ 0.5 \\ -0.056 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0.5 \\ 0.167 \\ 0 \\ 0.5 \\ -0.167 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$F = \begin{pmatrix} 0 \\ 0.5 \\ 0.056 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0.5 \\ -0.056 \end{pmatrix} \text{ kN}$$

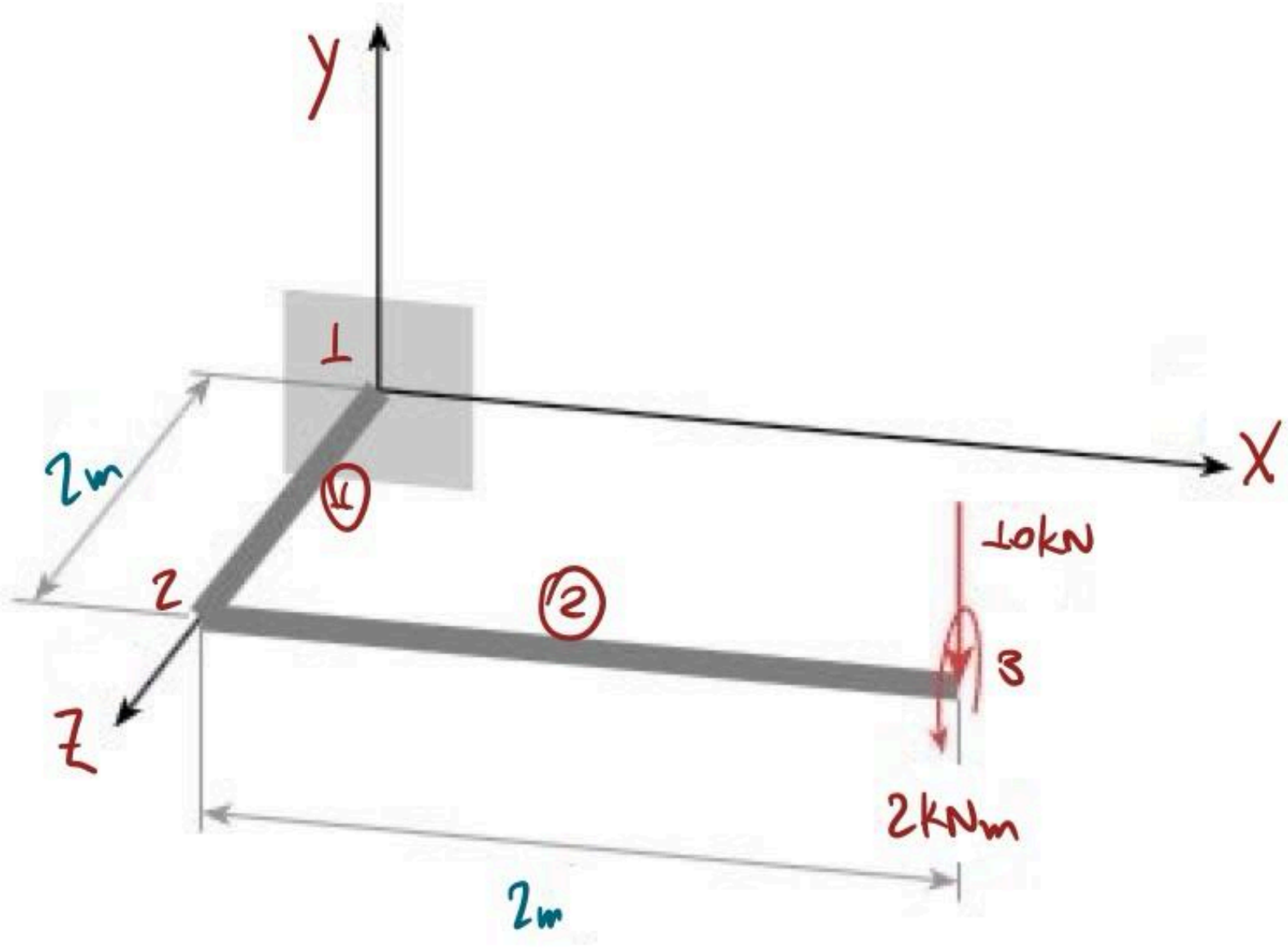
5.5. Resultados

Nó	u **(mm)**	v **(mm)**	θ **(rad)**	f **(kN)**	V **(kN)**	M **(kN · m)**
1	0	0	0	0	0.5	0.056
2	0.295	−0.001	−0.295	0	0	0
3	0.295	−0.001	0.295	0	0	0
4	0	0	0	0	0.5	−0.056

Tabela 6: Deslocamentos e esforços nodais.

6. Exercício V

Para este exercício, usou-se a formulação descrita pelo livro “A First Course in the Finite Element Method”, do autor Daryl L. Logan. Nesse livro, a formulação é realizada com o eixo y para cima. Assim, foi utilizada a convenção do livro, bem como a seguinte numeração de nós e elementos:



Segundo o autor, a matriz de rigidez de um pórtico espacial é apresentada abaixo:

$$\bar{K} = \left(\begin{array}{c|cccccc|cccccc} & u_{1x} & u_{1y} & u_{1z} & \theta_{1x} & \theta_{1y} & \theta_{1z} & u_{2x} & u_{2y} & u_{2z} & \theta_{2x} & \theta_{2y} & \theta_{2z} \\ f_{1x} & \frac{EA}{l} & 0 & 0 & 0 & 0 & 0 & -\frac{EA}{l} & 0 & 0 & 0 & 0 & 0 \\ f_{1y} & 0 & \frac{12EI_z}{l^3} & 0 & 0 & 0 & \frac{6EI_z}{l^2} & 0 & -\frac{12EI_z}{l^3} & 0 & 0 & 0 & \frac{6EI_z}{l^2} \\ f_{1z} & 0 & 0 & \frac{12EI_y}{l^3} & 0 & -\frac{6EI_y}{l^2} & 0 & 0 & 0 & -\frac{12EI_y}{l^3} & 0 & -\frac{6EI_y}{l^2} & 0 \\ M_{1x} & 0 & 0 & 0 & \frac{GJ}{l} & 0 & 0 & 0 & 0 & 0 & -\frac{GJ}{l} & 0 & 0 \\ M_{1y} & 0 & 0 & -\frac{6EI_y}{l^2} & 0 & \frac{4EI_y}{l} & 0 & 0 & 0 & \frac{6EI_y}{l^2} & 0 & \frac{2EI_y}{l} & 0 \\ M_{1z} & 0 & \frac{6EI_z}{l^2} & 0 & 0 & 0 & \frac{4EI_z}{l} & 0 & -\frac{6EI_z}{l^2} & 0 & 0 & 0 & \frac{2EI_z}{l} \\ \hline f_{2x} & -\frac{EA}{l} & 0 & 0 & 0 & 0 & 0 & \frac{EA}{l} & 0 & 0 & 0 & 0 & 0 \\ f_{2y} & 0 & -\frac{12EI_z}{l^3} & 0 & 0 & 0 & -\frac{6EI_z}{l^2} & 0 & \frac{12EI_z}{l^3} & 0 & 0 & 0 & -\frac{6EI_z}{l^2} \\ f_{2z} & 0 & 0 & -\frac{12EI_y}{l^3} & 0 & \frac{6EI_y}{l^2} & 0 & 0 & 0 & \frac{12EI_y}{l^3} & 0 & \frac{6EI_y}{l^2} & 0 \\ M_{2x} & 0 & 0 & 0 & -\frac{GJ}{l} & 0 & 0 & 0 & 0 & 0 & \frac{GJ}{l} & 0 & 0 \\ M_{2y} & 0 & 0 & -\frac{6EI_y}{l^2} & 0 & \frac{2EI_y}{l} & 0 & 0 & 0 & \frac{6EI_y}{l^2} & 0 & \frac{4EI_y}{l} & 0 \\ M_{2z} & 0 & \frac{6EI_z}{l^2} & 0 & 0 & 0 & \frac{2EI_z}{l} & 0 & -\frac{6EI_z}{l^2} & 0 & 0 & 0 & \frac{4EI_z}{l} \end{array}\right)$$

O módulo de cisalhamento é:

$$G = \frac{E}{2(1+\nu)} = \frac{180 \cdot 10^6}{2(1+0.3)}$$

$$G = 69.23 \cdot 10^6 \text{kN/m}^2$$

$\frac{EA}{l}$	$\frac{180 \cdot 10^6 \cdot 5 \cdot 10^{-3}}{2}$	$450 \cdot 10^3$ kN/m
$\frac{12EI}{l^3}$	$\frac{12 \cdot 378}{2^3}$	567 kN/m
$\frac{6EI}{l^2}$	$\frac{6 \cdot 378}{2^2}$	567 kN
$\frac{4EI}{l}$	$\frac{4 \cdot 378}{2}$	756 kN · m
$\frac{2EI}{l}$	$\frac{2 \cdot 378}{2}$	378 kN · m
$\frac{GJ}{l}$	$\frac{69.23 \cdot 10^6 \cdot 3.5 \cdot 10^{-6}}{2}$	121.15 kN · m

Tabela 7: Coeficientes da matriz de rigidez local.

Portanto, a matriz local de rigidez dos elementos é:

$$\bar{K} = 10^3 \left(\begin{array}{cccccc|cccccc} 450 & 0 & 0 & 0 & 0 & 0 & -450 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.567 & 0 & 0 & 0 & 0.567 & 0 & -0.567 & 0 & 0 & 0 & 0.567 \\ 0 & 0 & 0.567 & 0 & -0.567 & 0 & 0 & 0 & -0.567 & 0 & -0.567 & 0 \\ 0 & 0 & 0 & 0.12115 & 0 & 0 & 0 & 0 & 0 & -0.12115 & 0 & 0 \\ 0 & 0 & -0.567 & 0 & 0.756 & 0 & 0 & 0 & 0.567 & 0 & 0.378 & 0 \\ 0 & 0.567 & 0 & 0 & 0 & 0.756 & 0 & -0.567 & 0 & 0 & 0 & 0.378 \\ \hline -450 & 0 & 0 & 0 & 0 & 0 & 450 & 0 & 0 & 0 & 0 & 0 \\ 0 & -0.567 & 0 & 0 & 0 & -0.567 & 0 & 0.567 & 0 & 0 & 0 & -0.567 \\ 0 & 0 & -0.567 & 0 & 0.567 & 0 & 0 & 0 & 0.567 & 0 & 0.567 & 0 \\ 0 & 0 & 0 & -0.12115 & 0 & 0 & 0 & 0 & 0 & 0.12115 & 0 & 0 \\ 0 & 0 & -0.567 & 0 & 0.378 & 0 & 0 & 0 & 0.567 & 0 & 0.756 & 0 \\ 0 & 0.567 & 0 & 0 & 0 & 0.378 & 0 & -0.567 & 0 & 0 & 0 & 0.756 \end{array}\right)$$

As matrizes de transformação T podem ser expressas como:

$$T = \begin{pmatrix} \lambda & 0 & 0 & 0 \\ 0 & \lambda & 0 & 0 \\ 0 & 0 & \lambda & 0 \\ 0 & 0 & 0 & \lambda \end{pmatrix}$$

Em que:

$$\lambda = \begin{pmatrix} l & m & n \\ -\frac{m}{D} & \frac{l}{D} & 0 \\ -\frac{ln}{D} & -\frac{mn}{D} & D \end{pmatrix}$$

$$D = \sqrt{l^2 + m^2}$$

Em que l, m e n são os valores dos cossenos diretores em relação ao eixo $\bar{x}$ (local da barra).

- Elemento 1

O livro do Logan define uma exceção para a regra acima, quando o eixo $\bar{x}$ da barra coincide com o eixo Z global, o que gera um D igual a 0. Quando a barra aponta no mesmo sentido que o vetor Z, a matriz λ é definida por:

$$\lambda = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$$

Dessa forma:

$$T = \left(\begin{array}{ccc|ccc|ccc|ccc}
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0
\end{array}\right)$$

$$K = T^T \bar{K} T$$

$$\left(\begin{array}{ccc|ccc|ccc|ccc}
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0
\end{array}\right)^T \cdot \left(\begin{array}{cccccc|cccccc}
450 & 0 & 0 & 0 & 0 & 0 & -450 & 0 & 0 & 0 & 0 & 0 \\
0 & 0.567 & 0 & 0 & 0 & 0.567 & 0 & -0.567 & 0 & 0 & 0 & 0.567 \\
0 & 0 & 0.567 & 0 & -0.567 & 0 & 0 & 0 & -0.567 & 0 & -0.567 & 0 \\
0 & 0 & 0 & 0.12115 & 0 & 0 & 0 & 0 & 0 & -0.12115 & 0 & 0 \\
0 & 0 & -0.567 & 0 & 0.756 & 0 & 0 & 0 & 0.567 & 0 & 0.378 & 0 \\
0 & 0.567 & 0 & 0 & 0 & 0.756 & 0 & -0.567 & 0 & 0 & 0 & 0.378 \\
\hline
-450 & 0 & 0 & 0 & 0 & 0 & 450 & 0 & 0 & 0 & 0 & 0 \\
0 & -0.567 & 0 & 0 & 0 & -0.567 & 0 & 0.567 & 0 & 0 & 0 & -0.567 \\
0 & 0 & -0.567 & 0 & 0.567 & 0 & 0 & 0 & 0.567 & 0 & 0.567 & 0 \\
0 & 0 & 0 & -0.12115 & 0 & 0 & 0 & 0 & 0 & 0.12115 & 0 & 0 \\
0 & 0 & -0.567 & 0 & 0.378 & 0 & 0 & 0 & 0.567 & 0 & 0.756 & 0 \\
0 & 0.567 & 0 & 0 & 0 & 0.378 & 0 & -0.567 & 0 & 0 & 0 & 0.756
\end{array}\right) \cdot \left(\begin{array}{ccc|ccc|ccc|ccc}
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0
\end{array}\right)$$

$$K_1 = 10^3 \left(\begin{array}{cccccc|cccccc}
0.567 & 0 & 0 & 0 & 0.567 & 0 & -0.567 & 0 & 0 & 0 & 0.567 & 0 \\
0 & 0.567 & 0 & -0.567 & 0 & 0 & 0 & -0.567 & 0 & -0.567 & 0 & 0 \\
0 & 0 & 450 & 0 & 0 & 0 & 0 & 0 & -450 & 0 & 0 & 0 \\
0 & -0.567 & 0 & 0.756 & 0 & 0 & 0 & 0.567 & 0 & 0.378 & 0 & 0 \\
0.567 & 0 & 0 & 0 & 0.756 & 0 & -0.567 & 0 & 0 & 0 & 0.378 & 0 \\
0 & 0 & 0 & 0 & 0 & 0.12115 & 0 & 0 & 0 & 0 & 0 & -0.12115 \\
\hline
-0.567 & 0 & 0 & 0 & -0.567 & 0 & 0.567 & 0 & 0 & 0 & -0.567 & 0 \\
0 & -0.567 & 0 & 0.567 & 0 & 0 & 0 & 0.567 & 0 & 0.567 & 0 & 0 \\
0 & 0 & -450 & 0 & 0 & 0 & 0 & 0 & 450 & 0 & 0 & 0 \\
0 & -0.567 & 0 & 0.378 & 0 & 0 & 0 & 0.567 & 0 & 0.756 & 0 & 0 \\
0.567 & 0 & 0 & 0 & 0.378 & 0 & -0.567 & 0 & 0 & 0 & 0.756 & 0 \\
0 & 0 & 0 & 0 & 0 & -0.12115 & 0 & 0 & 0 & 0 & 0 & 0.12115
\end{array}\right)$$

- Elemento 2

Utilizando-se dos cossenos diretores:

$$\cos(\theta)_{x\bar{x}} = \frac{x_2 - x_1}{L} = \frac{2-0}{2}$$

$$\cos(\theta)_{x\bar{x}} = l = 1$$

$$\cos(\theta)_{y\bar{x}} = \frac{y_2 - y_1}{L} = \frac{0-0}{2}$$

$$\cos(\theta)_{y\bar{x}} = m = 0$$

$$\cos(\theta)_{z\bar{x}} = \frac{z_2 - z_1}{L} = \frac{0-0}{2}$$

$$\cos(\theta)_{z\bar{x}} = n = 0$$

Portanto,

$$D = \sqrt{1^2 + 0^2}$$

$$D = 1$$

$$\lambda = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Dessa forma:

$$T = \left(\begin{array}{ccc|ccc|ccc|ccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right)$$

Percebe-se que isso gera uma matriz identidade, o que permite afirmar que as matrizes de rigidez global e local do elemento 2 são iguais.

$$K_2 = 10^3 \left(\begin{array}{cccccc|cccccc}
450 & 0 & 0 & 0 & 0 & 0 & -450 & 0 & 0 & 0 & 0 & 0 \\
0 & 0.567 & 0 & 0 & 0 & 0.567 & 0 & -0.567 & 0 & 0 & 0 & 0.567 \\
0 & 0 & 0.567 & 0 & -0.567 & 0 & 0 & 0 & -0.567 & 0 & -0.567 & 0 \\
0 & 0 & 0 & 0.12115 & 0 & 0 & 0 & 0 & 0 & -0.12115 & 0 & 0 \\
0 & 0 & -0.567 & 0 & 0.756 & 0 & 0 & 0 & 0.567 & 0 & 0.378 & 0 \\
0 & 0.567 & 0 & 0 & 0 & 0.756 & 0 & -0.567 & 0 & 0 & 0 & 0.378 \\
\hline
-450 & 0 & 0 & 0 & 0 & 0 & 450 & 0 & 0 & 0 & 0 & 0 \\
0 & -0.567 & 0 & 0 & 0 & -0.567 & 0 & 0.567 & 0 & 0 & 0 & -0.567 \\
0 & 0 & -0.567 & 0 & 0.567 & 0 & 0 & 0 & 0.567 & 0 & 0.567 & 0 \\
0 & 0 & 0 & -0.12115 & 0 & 0 & 0 & 0 & 0 & 0.12115 & 0 & 0 \\
0 & 0 & -0.567 & 0 & 0.378 & 0 & 0 & 0 & 0.567 & 0 & 0.756 & 0 \\
0 & 0.567 & 0 & 0 & 0 & 0.378 & 0 & -0.567 & 0 & 0 & 0 & 0.756
\end{array}\right)$$

6.1. Matriz de rigidez global

$$K = \left(\begin{array}{c|cccccc|cccccc|cccccc}
 & u_{1x} & u_{1y} & u_{1z} & \theta_{1x} & \theta_{1y} & \theta_{1z} & u_{2x} & u_{2y} & u_{2z} & \theta_{2x} & \theta_{2y} & \theta_{2z} & u_{3x} & u_{3y} & u_{3z} & \theta_{3x} & \theta_{3y} & \theta_{3z} \\
\hline
f_{1x} & (0.567) & () & () & () & (0.567) & () & (-0.567) & () & () & () & (0.567) & () & () & () & () & () & () & () \\
f_{1y} & () & (0.567) & () & (-0.567) & () & () & () & (-0.567) & () & (-0.567) & () & () & () & () & () & () & () & () \\
f_{1z} & () & () & (450) & () & () & () & () & () & (-450) & () & () & () & () & () & () & () & () & () \\
M_{1x} & () & (-0.567) & () & (0.756) & () & () & () & (0.567) & () & (0.378) & () & () & () & () & () & () & () & () \\
M_{1y} & (0.567) & () & () & () & (0.756) & () & (-0.567) & () & () & () & (0.378) & () & () & () & () & () & () & () \\
M_{1z} & () & () & () & () & () & (0.12115) & () & () & () & () & () & (-0.12115) & () & () & () & () & () & () \\
\hline
f_{2x} & (-0.567) & () & () & () & (-0.567) & () & (0.567+450) & () & () & () & (-0.567) & () & (-450) & () & () & () & () & () \\
f_{2y} & () & (-0.567) & () & (0.567) & () & () & () & (0.567+0.567) & () & (0.567) & () & (0.567) & () & (-0.567) & () & () & () & (0.567) \\
f_{2z} & () & () & (-450) & () & () & () & () & () & (450+0.567) & () & (-0.567) & () & () & () & (-0.567) & () & (-0.567) & () \\
M_{2x} & () & (-0.567) & () & (0.378) & () & () & () & (0.567) & () & (0.756+0.12115) & () & () & () & () & () & (-0.12115) & () & () \\
M_{2y} & (0.567) & () & () & () & (0.378) & () & (-0.567) & () & (-0.567) & () & (0.756+0.756) & () & () & () & (0.567) & () & (0.378) & () \\
M_{2z} & () & () & () & () & () & (-0.12115) & () & (0.567) & () & () & () & (0.12115+0.756) & () & (-0.567) & () & () & () & (0.378) \\
\hline
f_{3x} & () & () & () & () & () & () & (-450) & () & () & () & () & () & (450) & () & () & () & () & () \\
f_{3y} & () & () & () & () & () & () & () & (-0.567) & () & () & () & (-0.567) & () & (0.567) & () & () & () & (-0.567) \\
f_{3z} & () & () & () & () & () & () & () & () & (-0.567) & () & (0.567) & () & () & () & (0.567) & () & (0.567) & () \\
M_{3x} & () & () & () & () & () & () & () & () & () & (-0.12115) & () & () & () & () & () & (0.12115) & () & () \\
M_{3y} & () & () & () & () & () & () & () & () & (-0.567) & () & (0.378) & () & () & () & (0.567) & () & (0.756) & () \\
M_{3z} & () & () & () & () & () & () & () & (0.567) & () & () & () & (0.378) & () & (-0.567) & () & () & () & (0.756)
\end{array}\right)$$

$$K = 10^3 \begin{pmatrix}
0.567 & 0 & 0 & 0 & 0.567 & 0 & -0.567 & 0 & 0 & 0 & 0.567 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0.567 & 0 & -0.567 & 0 & 0 & 0 & -0.567 & 0 & -0.567 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 450 & 0 & 0 & 0 & 0 & 0 & -450 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -0.567 & 0 & 0.756 & 0 & 0 & 0 & 0.567 & 0 & 0.378 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0.567 & 0 & 0 & 0 & 0.756 & 0 & -0.567 & 0 & 0 & 0 & 0.378 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0.12115 & 0 & 0 & 0 & 0 & 0 & -0.12115 & 0 & 0 & 0 & 0 & 0 & 0 \\
-0.567 & 0 & 0 & 0 & -0.567 & 0 & 450.567 & 0 & 0 & 0 & -0.567 & 0 & -450 & 0 & 0 & 0 & 0 & 0 \\
0 & -0.567 & 0 & 0.567 & 0 & 0 & 0 & 1.134 & 0 & 0.567 & 0 & 0.567 & 0 & -0.567 & 0 & 0 & 0 & 0.567 \\
0 & 0 & -450 & 0 & 0 & 0 & 0 & 0 & 450.567 & 0 & -0.567 & 0 & 0 & 0 & -0.567 & 0 & -0.567 & 0 \\
0 & -0.567 & 0 & 0.378 & 0 & 0 & 0 & 0.567 & 0 & 0.87715 & 0 & 0 & 0 & 0 & 0 & -0.12115 & 0 & 0 \\
0.567 & 0 & 0 & 0 & 0.378 & 0 & -0.567 & 0 & -0.567 & 0 & 1.512 & 0 & 0 & 0 & 0.567 & 0 & 0.378 & 0 \\
0 & 0 & 0 & 0 & 0 & -0.12115 & 0 & 0.567 & 0 & 0 & 0 & 0.87715 & 0 & -0.567 & 0 & 0 & 0 & 0.378 \\
0 & 0 & 0 & 0 & 0 & 0 & -450 & 0 & 0 & 0 & 0 & 0 & 450 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -0.567 & 0 & 0 & 0 & -0.567 & 0 & 0.567 & 0 & 0 & 0 & -0.567 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0.567 & 0 & 0.567 & 0 & 0 & 0 & 0.567 & 0 & 0.567 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0.12115 & 0 & 0 & 0 & 0 & 0 & 0.12115 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0.567 & 0 & 0.378 & 0 & 0 & 0 & 0.567 & 0 & 0.756 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.567 & 0 & 0 & 0 & 0.378 & 0 & -0.567 & 0 & 0 & 0 & 0.756
\end{pmatrix}$$

6.2. Vetores Força e Deslocamentos

$$F = \begin{pmatrix} f_{1x} \\ f_{1y} \\ f_{1z} \\ M_{1x} \\ M_{1y} \\ M_{1z} \\ f_{2x} \\ f_{2y} \\ f_{2z} \\ M_{2x} \\ M_{2y} \\ M_{2z} \\ f_{3x} \\ f_{3y} \\ f_{3z} \\ M_{3x} \\ M_{3y} \\ M_{3z} \end{pmatrix} = \begin{pmatrix} f_{1x} \\ f_{1y} \\ f_{1z} \\ M_{1x} \\ M_{1y} \\ M_{1z} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -10 \\ 0 \\ 2 \\ 0 \\ 0 \end{pmatrix}$$

$$U = \begin{pmatrix} u_{1x} \\ u_{1y} \\ u_{1z} \\ \theta_{1x} \\ \theta_{1y} \\ \theta_{1z} \\ u_{2x} \\ u_{2y} \\ u_{2z} \\ \theta_{2x} \\ \theta_{2y} \\ \theta_{2z} \\ u_{3x} \\ u_{3y} \\ u_{3z} \\ \theta_{3x} \\ \theta_{3y} \\ \theta_{3z} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ u_{2x} \\ u_{2y} \\ u_{2z} \\ \theta_{2x} \\ \theta_{2y} \\ \theta_{2z} \\ u_{3x} \\ u_{3y} \\ u_{3z} \\ \theta_{3x} \\ \theta_{3y} \\ \theta_{3z} \end{pmatrix}$$

6.3. Resolução do Sistema Linear

Aplicando o método da eliminação:

$$KU = F$$

$$\begin{pmatrix}
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 450.567 & 0 & 0 & 0 & -0.567 & 0 & -450 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1.134 & 0 & 0.567 & 0 & 0.567 & 0 & -0.567 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 450.567 & 0 & -0.567 & 0 & 0 & 0 & -0.567 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.567 & 0 & 0.87715 & 0 & 0 & 0 & 0 & 0 & -0.12115 \\
0 & 0 & 0 & 0 & 0 & 0 & -0.567 & 0 & -0.567 & 0 & 1.512 & 0 & 0 & 0 & 0.567 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.567 & 0 & 0 & 0 & 0.87715 & 0 & -0.567 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -450 & 0 & 0 & 0 & 0 & 0 & 450 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -0.567 & 0 & 0 & 0 & -0.567 & 0 & 0.567 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0.567 & 0 & 0.567 & 0 & 0 & 0 & 0.567 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0.12115 & 0 & 0 & 0 & 0 & 0 & 0.12115 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0.567 & 0 & 0.378 & 0 & 0 & 0 & 0.567 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.567 & 0 & 0 & 0 & 0.378 & 0 & -0.567 & 0 & 0
\end{pmatrix}
\cdot
\begin{pmatrix}
u_{1x} \\ u_{1y} \\ u_{1z} \\ \theta_{1x} \\ \theta_{1y} \\ \theta_{1z} \\ u_{2x} \\ u_{2y} \\ u_{2z} \\ \theta_{2x} \\ \theta_{2y} \\ \theta_{2z} \\ u_{3x} \\ u_{3y} \\ u_{3z} \\ \theta_{3x} \\ \theta_{3y} \\ \theta_{3z}
\end{pmatrix}
=
\begin{pmatrix}
0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -10 \\ 0 \\ 2 \\ 0 \\ 0
\end{pmatrix}$$

6.4. Resultados

$$U = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -81.129 \\ 0 \\ 63.492 \\ 0 \\ -165.085 \\ 0 \\ -481.845 \\ 0 \\ 80.001 \\ 0 \\ -217.995 \end{pmatrix} \text{ mm, rad}$$

$$F = KU$$

$$F = \begin{pmatrix} 0 \\ 10 \\ 0 \\ -22 \\ 0 \\ 20 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -10 \\ 0 \\ 2 \\ 0 \\ 0 \end{pmatrix} \text{ kN, kN·m}$$

Nó	GDL	U (mm, rad)	Força (kN, kN·m)
1	u_x	0	0
1	u_y	0	10
1	u_z	0	0
1	θ_x	0	−22
1	θ_y	0	0
1	θ_z	0	20
2	u_x	0	0
2	u_y	−81.129	0
2	u_z	0	0
2	θ_x	63.492	0
2	θ_y	0	0
2	θ_z	−165.085	0
3	u_x	0	0
3	u_y	−481.845	−10
3	u_z	0	0
3	θ_x	80.001	2
3	θ_y	0	0
3	θ_z	−217.995	0